

NETWORK CITY

CLOUD, SERVERS & VIRTUALIZATION

Serve. Share. Scale.



LEARN THE ROLES. SHARE RESOURCES. SOLVE THE SCENARIO.

Detailed comic illustrations + short definitions + clear exam clues

Explore server jobs, cloud service models, and virtual machines. Learn objectives 2.3, 4.1, and 4.2. Practice 24 flashcards in the matching clickable PowerPoint.

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Independent study guide. Aligned to exam 220-1201; not an official CompTIA publication. Illustrations are memory aids. Technical notes and labeled diagrams explain the real rules.

Three heroes, one connected world

VISUAL: Draw a smiling cloud above a real data center. Neon cyan server heroes and a purple VM wizard help a student.



Serve a job. Share the hardware. Rent a service!

WOOSH!

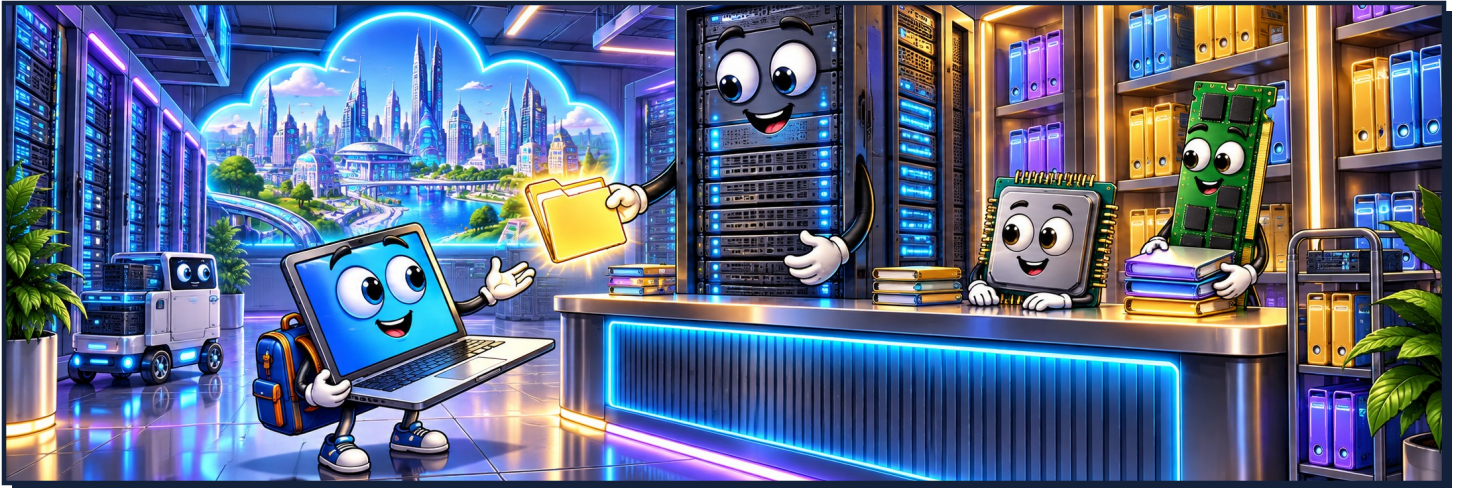
- **Server.** A computer or program that provides a service to other devices. A client asks for that service. Your browser is a client when it asks a web server for a page.
- **Virtualization.** Software creates separate computing environments from physical resources. One real computer can run several virtual computers, each with its own work.
- **Cloud computing.** Computing resources become services you can request through a network. Real machines in real buildings still do the work. The cloud is not floating storage in the sky.

- **How they connect.** A cloud provider may use virtual machines on physical servers to run your app. A server can also run at your school without being a cloud service.
- **Our comic rule.** The characters help memory. The bold definitions explain the real technology. Extra lab tips build understanding but are not a promise of an exact exam question.

EXAM MAP: 220-1201 objective 2.3 covers host services; 4.1 covers virtualization; 4.2 covers cloud concepts. Related port details support 2.1. Independent study material. [1,2]

Servers: the service crew

VISUAL: Draw a googly-eyed server librarian in a glowing rack. Add gold CPU and memory helpers, vents, rails, and blue lights.



Ask for help. The server does its job!

SERVE!

- **Client-server.** A client sends a request. A server responds. One server can help many clients. A computer can act as a client for one task and a server for another.

- **Role / service.** A role is a job, such as sharing files. A service is a program that performs work, often in the background. One machine can run several roles if its resources and security plan allow it.

- **Host / endpoint.** A host is a network-connected system. An endpoint is a device at an end of communication. In virtualization, host also means the system that supplies a VM's physical resources.

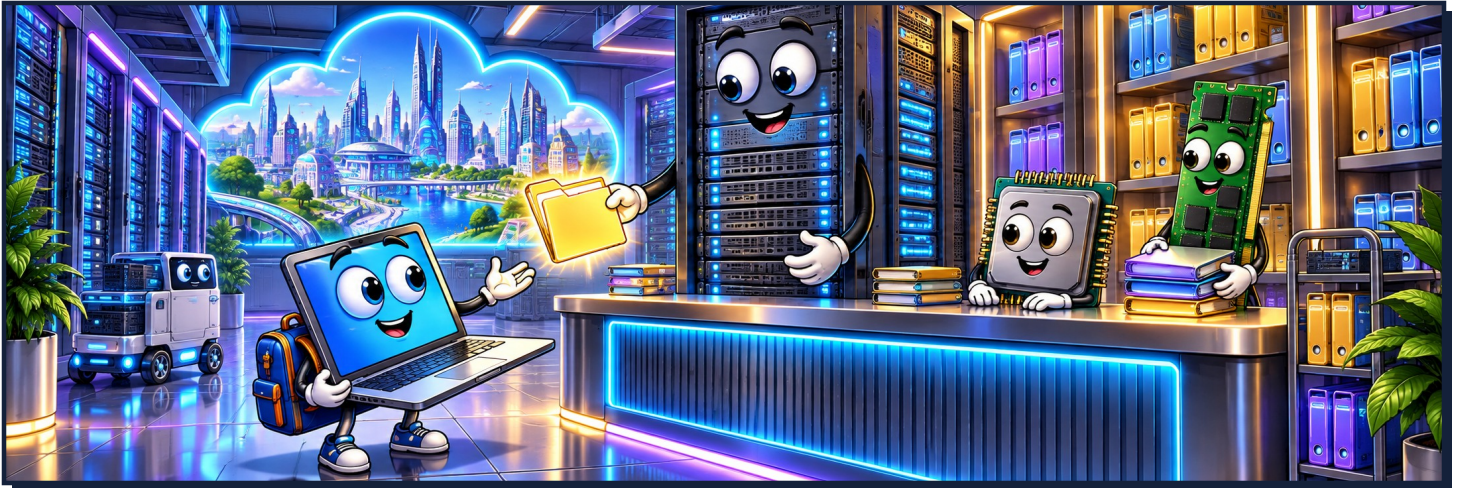
- **Physical versus virtual server.** A physical server is hardware. A virtual server runs in a software-created computer. Both can offer DNS, web pages, files, or other services.

- **On-premises / data center.** On-premises means at an organization's own site. A data center is a facility for computing equipment, power, cooling, and networking.

SCENARIO: A tiny office computer shares a folder with five coworkers. It performs a server role even if it looks like a desktop. "Server" describes the job as well as a hardware category. [1]

Server hardware & safe rack space

VISUAL: Draw a googly-eyed server librarian in a glowing rack. Add gold CPU and memory helpers, vents, rails, and blue lights.



Strong parts need air, power, and room!

COOL!

- **CPU / RAM.** Central Processing Unit does calculations. Random Access Memory holds active work. More users can need more CPU time and RAM. RAM is not long-term file storage.

- **Storage / NIC.** An SSD, Solid-State Drive, stores data without spinning platters. An HDD, Hard Disk Drive, uses spinning platters. A NIC, Network Interface Card, connects the server to a network.

- **Rack unit: U.** One U equals 1.75 inches of vertical rack space. A 2U device uses 3.5 inches. Rack depth, rails, weight, and cooling space still must match the hardware. [15]

- **UPS / redundant power.** An Uninterruptible Power Supply provides temporary battery power. Redundant power supplies can reduce failures when correctly connected. Neither gives unlimited runtime.

- **Safe handling.** Follow the maker's rack and lifting instructions. Keep heavy gear low, stabilize the rack, and protect airflow. Shut down and unplug before service unless a supported hot-swap procedure says otherwise.

EXTRA HARDWARE CONTEXT: Use ESD (Electrostatic Discharge) precautions for sensitive parts. Never open a power supply. A server's safe capacity depends on its exact model and workload; there is no universal speed or clearance. [1,15]

DNS: the name-finding librarian

VISUAL: Draw a neon service hall. A DNS librarian, address-lease clerk, file robot, and web-page helper welcome laptops.



Give me a name. I help find its address!

LOOK UP!

- **DNS - Domain Name System.** A distributed naming system. One common job maps a host name to an IP address. IP means Internet Protocol. A name is easier to remember than a long address.

- **Query / resolver.** A query is a question. A resolver finds an answer for the client. It may use a saved answer or ask other DNS servers.

- **Record / cache.** A record stores a piece of DNS information. A cache keeps an answer for reuse. An A record holds an Internet Protocol version 4 (IPv4) address. An AAAA record holds a version 6 (IPv6) address.

- **Other records.** CNAME means Canonical Name: an alias points to another name. MX means Mail Exchanger: it identifies mail servers for a domain. These are useful naming terms, beyond the basic role.

- **Port clue.** DNS commonly uses UDP or TCP port 53. UDP means User Datagram Protocol. TCP means Transmission Control Protocol. A port number identifies a service endpoint in network traffic.

EXAM CLUE: A resource works by IP address but fails by name? Investigate DNS settings and name resolution. This clue suggests a cause; it does not prove one. DNS does not lease client IP settings. [1,6]

DHCP: the address-lease clerk

●●● **VISUAL:** Draw a neon service hall. A DNS librarian, address-lease clerk, file robot, and web-page helper welcome laptops.



Here is your address—and the rules to use it!

LEASE!

- **DHCP - Dynamic Host Configuration Protocol.**

Automatically gives clients network settings. A lease lets a client use an address for a period of time. A client normally renews its lease.

- **Scope / pool.** The range of addresses and settings a DHCP server can offer for a network. It must not hand the same active address to two clients.

- **Options.** Extra settings can include subnet mask, default gateway, and DNS server addresses. The mask helps identify the local network. The gateway leads toward other networks.

- **Reservation / static address.** A reservation makes DHCP offer a chosen address to a matching client. A static address is manually configured. Plan both to avoid conflicts.

- **IPv4 exchange.** Discover, Offer, Request, Acknowledge: DORA. The client looks for help, the server offers, the client requests, and the server confirms. DHCPv4 uses UDP server port 67 and client port 68.

MEMORY HOOK: DHCP = settings desk. DNS = name desk. A Windows IPv4 address beginning 169.254 can suggest automatic link-local addressing after DHCP fails. Check the link and DHCP path. [1,7]

File & print servers: shared helpers

VISUAL: Draw a neon service hall. A DNS librarian, address-lease clerk, file robot, and web-page helper welcome laptops.



One shared place. Clear rules for every user!

SHARE!

- **File server.** Stores files and offers network access. A shared folder is a named location that allowed users can open. Central storage helps teams work from a common location.

- **SMB - Server Message Block.** A network protocol for sharing files and other resources. Modern direct-hosted SMB commonly uses TCP port 445. A protocol is a set of communication rules. [1,8]

- **Permissions.** Rules decide who can read, change, or remove data. A network connection alone does not grant access. Check both sharing rules and file-system permissions.

- **Print server.** Accepts jobs for shared printers. A queue is the waiting line. A spooler stores and schedules print data. One stuck job can delay others.

- **NAS - Network-Attached Storage.** A network device built mainly to provide storage services. It often offers file shares. A NAS can still fail, so important files need backups.

SCENARIO: Everyone can reach the file server, but one user cannot open a folder. Check the user's permissions before replacing a network cable. A shared drive is not automatically a backup. [1,8]

Web & database: show it, store it

●●● **VISUAL:** Draw a neon service hall. A DNS librarian, address-lease clerk, file robot, and web-page helper welcome laptops.



The web counter serves pages. The database keeps records!

FETCH!

- **Web server.** Handles requests for web content. A browser may request a page, image, or data for an app. Static content is already stored; dynamic content can be built when requested.

- **HTTP / HTTPS.** Hypertext Transfer Protocol carries web requests and responses. HTTPS is HTTP protected with TLS, Transport Layer Security. Learn ports 80 for HTTP and 443 for HTTPS. [1]

- **Database server.** Stores organized data and answers requests to find or change it. An online store might keep product names, prices, and stock counts in a database.

- **Query / SQL.** A query asks for data or an action. SQL means Structured Query Language. Many relational databases use it. A table groups data into rows and columns.

- **Separate jobs.** The web app handles the visitor's request and may ask the database for results. A database need not be directly open to visitors on the internet.

MEMORY HOOK: Web = the service counter. Database = the organized records behind it. HTTPS protects a connection; the padlock alone does not prove that a website is honest. [1,16]

Mail servers: the message route

VISUAL: Draw a neon traffic marshal sending request carts to two servers. Add a proxy desk and a shield-shaped mail gate.



Send, store, and read are different jobs!

SEND!

- **Mail server.** Receives, routes, and stores electronic mail. A mailbox is a user's message store. A client app or webmail page lets the user work with it.

- **SMTP - Simple Mail Transfer Protocol.** Moves outgoing mail and relays mail between servers. Core 1 names TCP port 25. Authenticated client submission often uses other ports configured by the provider.

- **IMAP - Internet Message Access Protocol.** Lets a client work with mail kept on a server. Folders and message state can stay in step across devices. Core port: 143.

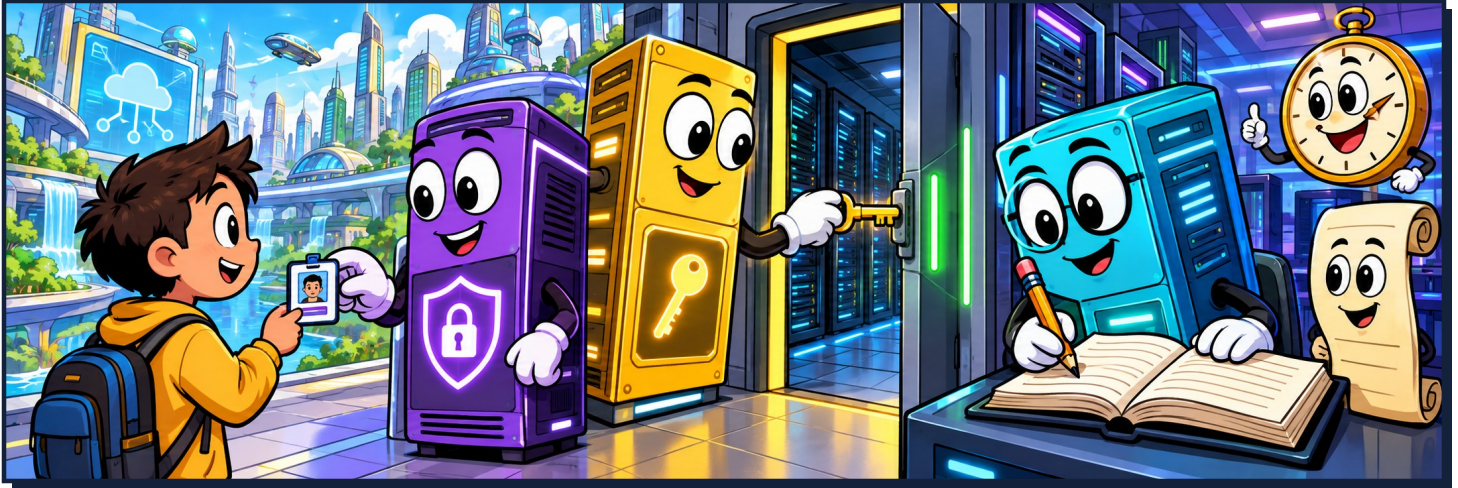
- **POP3 - Post Office Protocol version 3.** Retrieves messages from a server. Whether copies remain there depends on settings. Core port: 110. POP3 does not always mean "delete every server copy."

- **Secure connections.** Encryption settings must match the service. Ports 993 for IMAP over TLS and 995 for POP3 over TLS are useful extra context. A port number alone does not prove encryption.

SCENARIO: A user can read old messages but cannot send new ones. Check the outgoing-mail service, authentication, connection, and settings. Knowing the server's job helps narrow the problem. [1,17]

AAA: three guards at the gate

VISUAL: Draw three bright purple and cyan guards checking identity, opening an allowed door, and writing a record.



Who are you? What may you do? What happened?

CHECK!

- **AAA.** Authentication, Authorization, and Accounting. The three words describe related but different security jobs. [9]

- **Authentication.** Checks a claimed identity. A password or another approved sign-in method helps show who the user is. MFA means Multifactor Authentication: more than one factor type.

- **Authorization.** Decides what an identified user may do. A student may read a class folder. An instructor may also edit it. Successful sign-in does not mean unlimited access.

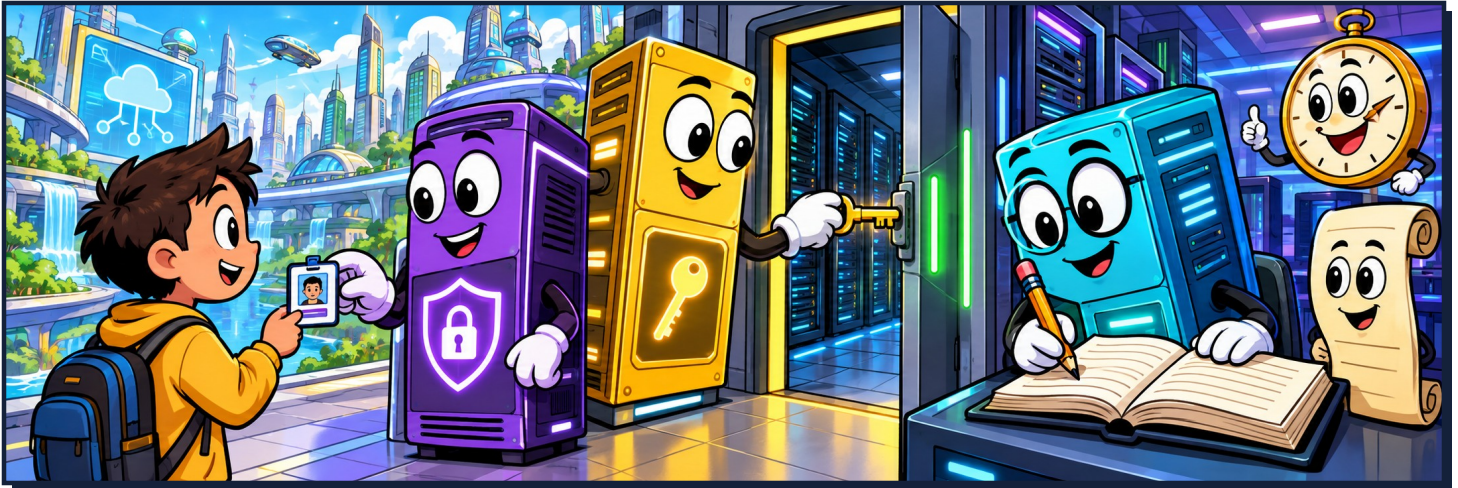
- **Accounting.** Records activity or resource use. Logs can show when someone connected or how long a session lasted. This is recordkeeping, not only money bookkeeping.

- **Least privilege.** Give people and services only the access needed for their work. Use a normal account for routine tasks and approved admin access for special changes.

MEMORY HOOK: Identity badge = authentication. Allowed door = authorization. Activity notebook = accounting. Do not swap the three A words on an exam scenario. [1,9]

Syslog & NTP: records need time

VISUAL: Draw three bright purple and cyan guards checking identity, opening an allowed door, and writing a record.



Write the event. Keep the clocks together!

TICK!

- **Syslog.** A standard way for systems to send event messages. A syslog server collects records from devices. “Syslog” is a name tied to system logging, not a three-letter acronym. [10]

- **Log / severity.** A log is a record of events. Severity describes how serious an event is. A central collector helps compare clues from several systems.

- **NTP - Network Time Protocol.** Helps synchronize device clocks over a network. “Synchronize” means keep things in step. NTP normally uses UDP port 123. [11]

- **Why time matters.** Accurate timestamps help place events in order. Large clock errors can also break some sign-in and certificate checks. Correct time does not fix the event itself.

- **Transport clue.** Traditional syslog often uses UDP 514. Other transports exist, including protected syslog over TLS. These port details add context; do not assume all logs travel the same way.

SCENARIO: Two logs seem to show a reply before its request. Compare clock settings and time zones before trusting the timeline. NTP sets the time; syslog carries the event messages. [1,10,11]

Network appliances: helpful middle stops

VISUAL: Draw a neon traffic marshal sending request carts to two servers. Add a proxy desk and a shield-shaped mail gate.



Sort the traffic. Share the work. Block the junk!

ROUTE!

- **Appliance.** A device or packaged virtual system built for a focused job. The same kinds of functions can also be offered as software or cloud services.

- **Spam gateway.** Filters unwanted or suspicious mail before it reaches users. Filtering reduces junk and threats, but users still need good judgment.

- **UTM - Unified Threat Management.** Combines security functions in one system. A product may combine a firewall, intrusion prevention, malware scanning, and content filtering. Features depend on the product.

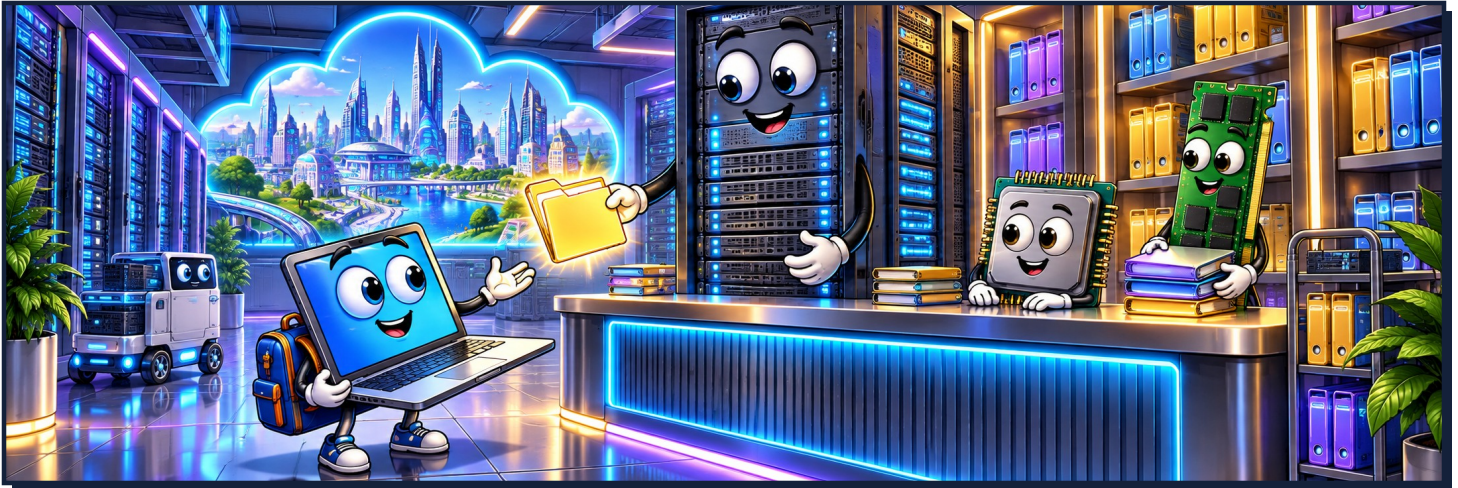
- **Load balancer.** Spreads requests among available servers. Health checks can avoid sending work to a failed server. It needs suitable configuration and enough healthy capacity. [12]

- **Proxy server.** An intermediary that makes or handles requests on behalf of another system. A forward proxy helps clients; a reverse proxy sits in front of servers. It may filter, cache, or relay traffic.

MEMORY HOOK: Load balancer = traffic marshal. Proxy = middle desk. Spam gateway = mail sorter. UTM = combined security checkpoint. These roles can overlap in a product. [1,12]

Embedded systems: small, focused jobs

VISUAL: Draw a googly-eyed server librarian in a glowing rack. Add gold CPU and memory helpers, vents, rails, and blue lights.



Tiny computer. Big real-world job!

SENSE!

- **Embedded system.** A computer built into another product for a focused purpose. It may have a small processor, limited memory, and special software.

- **IoT - Internet of Things.** Connected physical devices that sense, report, or act. Smart thermostats, sensors, and connected cameras are examples. They may use a gateway instead of talking straight to the internet.

- **SCADA.** Supervisory Control and Data Acquisition. A system gathers information from industrial equipment and supports monitoring and control. Think of water-pump or power-system operations. [18]

- **Legacy.** Older technology that remains in use. It may need old interfaces or unsupported software. Age alone does not tell you which changes are safe.

- **Safe support.** Inventory the devices, change default credentials, and separate access as policy requires. Use vendor-approved updates and maintenance windows. Ask the responsible operations team before disrupting control equipment.

EXAM CLUE: SCADA relates to monitoring and controlling physical processes. IoT relates to connected things. They can overlap, but the terms do not mean exactly the same thing. [1,18]

Cloud: a service, not a magic place

VISUAL: Draw three colorful rental workshops: computer building blocks, a ready tool bench, and a finished app counter.



Request what you need. Real hardware does the rest!

WOOSH!

- **On-demand self-service.** Users can request resources through an approved portal or tool without a person manually building every request. Access and spending rules still apply.
- **Broad network access.** Services can be reached through a network from supported devices. A private cloud may use a private network, not the public internet.

- **Resource pooling.** The provider combines resources and assigns them where needed. Like a shared fleet of rental cars, each customer does not need to buy the whole fleet.
- **Rapid elasticity.** Capacity can grow and shrink as demand changes. Cloud systems can make this fast. It is not a promise of unlimited capacity.
- **Measured service.** Usage is tracked, reported, or controlled. Some services charge by measured use; others use subscriptions or bundled plans.

NIST = National Institute of Standards and Technology. These five traits form its cloud definition. A remote file server alone does not automatically provide every cloud trait. [2]

IaaS, PaaS & SaaS: choose your job

VISUAL: Draw three colorful rental workshops: computer building blocks, a ready tool bench, and a finished app counter.



How much of the stack will your team manage?

CHOOSE!

Model	Full name	Typical customer work
IaaS	Infrastructure as a Service	Guest OS, installed apps, data, and access
PaaS	Platform as a Service	App code, data, and app configuration
SaaS	Software as a Service	Using/configuring the app; data and access

- Stack.** The layers that make an app work: hardware, operating system, runtime, and application. OS means Operating System. A runtime is software that runs an application's code.
- Control versus effort.** IaaS usually gives more system control and more management work. SaaS gives a finished app with less control of the underlying system.

MEMORY HOOK: IaaS = rent building blocks. PaaS = rent a ready workshop. SaaS = use the finished shop. Exact responsibilities depend on the service and agreement. [2,3,4]

IaaS: rent the computing blocks

VISUAL: Draw three colorful rental workshops: computer building blocks, a ready tool bench, and a finished app counter.



You manage the guest. We run the building!

BUILD!

- **What you get.** Infrastructure as a Service supplies resources such as virtual compute, network, and storage. You can build your own software environment on top.
- **Your work.** For a rented VM, you usually install and patch the guest OS, configure apps, control access, and protect your data. The provider manages the underlying physical infrastructure.

- **Why choose it.** Use it when an app needs OS-level control or a familiar server setup. An organization can move a server workload without buying a new physical machine.
- **Example mission.** A school rents a VM, installs a supported OS, patches it, and runs a practice website. Renting a VM does not make those guest-software tasks disappear.
- **Watch the whole bill.** Compute is only one item. Disks, backup, network transfer, and other services may add cost. Stopping a VM may leave billable storage behind.

EXAM CLUE: “We need control over the operating system” points toward IaaS. It does not mean the customer repairs the provider’s physical server. [2,3,5]

PaaS: bring your app to the workshop

VISUAL: Draw three colorful rental workshops: computer building blocks, a ready tool bench, and a finished app counter.



The tools are ready. Bring your code!

CREATE!

- **What you get.** Platform as a Service offers an environment for building or running applications. The provider handles the underlying infrastructure and supported platform pieces.
- **Your work.** You manage your app code, data, access choices, and settings within the service. You still fix bugs and protect secrets such as private access keys.

- **Runtime / framework.** A runtime executes code. A framework supplies reusable tools and patterns for building an app. A platform supports particular languages and versions.
- **Use case.** A development team deploys a web app to a managed app platform. It can focus on features instead of patching the platform's underlying OS.
- **Tradeoff.** The platform's supported features and limits matter. Moving an app to another platform can require changes. Do not assume every PaaS runs every program.

EXAM CLUE: "We will deploy our code while the provider manages the platform" points toward PaaS. The customer still owns app-level decisions and data responsibilities. [2,3,4]

SaaS: use the finished application

VISUAL: Draw three colorful rental workshops: computer building blocks, a ready tool bench, and a finished app counter.



Sign in and do the job!

READY!

- **What you get.** Software as a Service delivers a provider-run application. Users access it through supported software, often a web browser or mobile app.

- **Your work.** Set up users and allowed access. Choose sharing settings. Protect accounts and handle your organization's data. The provider runs the application and its underlying systems.

- **Use cases.** Hosted email, online document editing, and customer-management apps can be SaaS. Buying a downloadable desktop app alone does not automatically make it SaaS.

- **Subscription / tenant.** A subscription is an agreement for service use. A tenant is a customer's logically separated service environment. Several tenants may share underlying resources.

- **Check the service.** Ask about supported devices, offline use, data export, recovery, and account controls. A browser-based app can still have outages or require a license.

EXAM CLUE: "Users need a ready-made app with no server management" suggests SaaS. Security is still shared: protect identities, data, devices, and the settings you control. [2,4,5]

Four places a cloud can belong

●●● **VISUAL:** Draw a cyan public plaza, purple private garden, shared school garden, and gold bridge joining two cloud spaces.



Who uses this cloud—and how do clouds connect?

MAP!

Model	Who or what it serves	Memory picture
Public	Provider offers cloud services to many customers	Public rental plaza
Private	Exclusive use by one organization	One organization’s garden
Community	Organizations with shared needs or concerns	Shared school garden
Hybrid	Distinct clouds connected to work together	Bridge between cloud spaces

- **Private can be elsewhere.** A private cloud may be on-premises or hosted off-site. Ownership, management, and location can vary.
- **Public does not mean open data.** Public cloud describes service availability. Customer data still needs access controls; it is not meant to be readable by everyone.

EXAM TRAP: A private network alone is not a private cloud. A public cloud plus a private cloud linked for data or app use is a typical hybrid example. Community cloud serves shared organizational concerns. [2]

Shared, dedicated & multitenant

VISUAL: Draw a cyan public plaza, purple private garden, shared school garden, and gold bridge joining two cloud spaces.



Share the building. Keep the apartments separate!

POOL!

- **Shared resources.** Several customers use a common pool of capacity. Logical controls keep their work separated. Sharing hardware does not mean users can freely read each other's files.

- **Dedicated resources.** Capacity or hardware is reserved for a customer under the service's terms. It can give isolation or control but often costs more. Check what is actually dedicated.

- **Multitenancy.** A system serves multiple tenants with separation between them. Tenants may share an application, platform, or infrastructure while keeping distinct settings and data.

- **Private is a different question.** Private cloud describes exclusive organizational use. Dedicated hardware describes resource allocation. A public-cloud provider can offer dedicated resources.

- **Noisy neighbor.** Extra context: another workload can compete for shared capacity. Limits and resource controls help manage this. Sharing is useful, but it requires careful design.

MEMORY HOOK: Deployment model = who the cloud serves. Service model = what you manage. Resource model = what is shared or reserved. Ask each question separately. [2,4]

Scale & elasticity: handle the crowd

VISUAL: Draw a cheerful cloud manager adding server carts for a busy line. Park spare carts beside a glowing usage meter.



Add help for the rush. Give it back when quiet!

GROW!

- **Scalability.** The ability to handle more work by adding resources. Think of a store that can expand when its customer count grows.

- **Vertical scaling: up or down.** Change the resources in one system, such as adding RAM or CPU capacity. A bigger truck carries more in one trip.

- **Horizontal scaling: out or in.** Add or remove systems. More delivery trucks share the work. The application must be designed to use them well.

- **Elasticity.** Adjust capacity to match changing demand, including shrinking afterward. Cloud services may automate this with rules. Scaling once does not by itself prove elasticity.

- **Limits.** Quotas, budgets, start-up time, and app design still matter. A quota is an allowed limit. Extra servers cannot fix every database lock or slow network path.

SCENARIO: A site adds instances for a ticket sale, then removes them after the rush. That is elasticity. The load balancer can direct requests across the active servers. [2,12]

Metered use, speed & transfer

VISUAL: Draw a cheerful cloud manager adding server carts for a busy line. Park spare carts beside a glowing usage meter.



Watch the meter—and the road!

COUNT!

- **Metered utilization.** The provider measures resources consumed. Billing may include compute time, stored data, requests, and transfer. A subscription may also have included allowances.

- **Ingress / egress.** Ingress means data enters the named service or network. Egress means data leaves it. State the viewpoint. Downloading from a cloud service is usually egress from that service.

- **Costs vary.** Do not memorize “all ingress is free” or one price per gigabyte. Region, destination, service, and plan affect charges. Check the current rate card and billing units.

- **Bandwidth / latency.** Bandwidth is link capacity. Throughput is the useful transfer rate actually achieved. Latency is delay, often in ms: milliseconds. A fast link can still have a long wait.

- **Mbps / Gbps.** Megabits per second and gigabits per second measure bit rate. A byte has eight bits. Cloud service models have no fixed speed or cable-distance rating.

MEMORY HOOK: Bandwidth = road width. Throughput = cars that get through. Latency = travel delay. Watch budget alerts and unused resources as well as performance. [2,19]

Availability: keep the service reachable

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



A spare part helps only when the plan works!

STAY UP!

- **Availability.** How often a service can be used when needed. High availability uses design choices to reduce downtime. No design makes every possible failure disappear.

- **Redundancy / failover.** Redundancy adds extra components or copies. Failover moves work to another working resource after a failure. Test the process instead of assuming it works.

- **Region / zone.** A region is a geographic service area. An availability zone is a separate failure area within a region in providers that use this model. Exact designs differ.

- **SLA - Service Level Agreement.** Defines service commitments and conditions. An uptime target is not a promise that your whole application will never fail. Read the covered service, exclusions, and remedies.

- **Availability versus backup.** Live replicas can repeat accidental deletion or corruption. Backups preserve recoverable copies. A backup also needs a tested restore process.

EXAM CLUE: Several servers can improve availability when traffic, data, and failover are designed for it. One shared power source or one failed database can still stop the service. [2,20]

File sync: keep copies in step

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Sync follows changes. Recovery needs a plan!

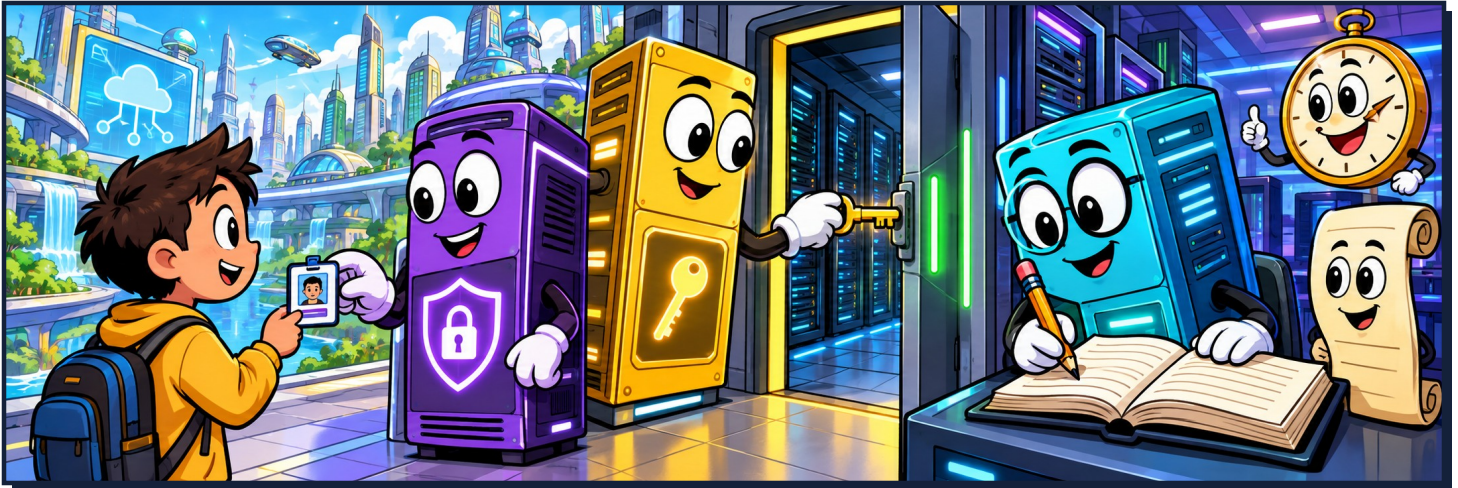
SYNC!

- **File synchronization.** Keeps selected files aligned across locations or devices. A sync client watches changes and exchanges them with the service.
- **Online-only / offline.** An online-only item may show a name without a full local copy. Download it or mark it for offline use as supported before losing internet access.
- **Conflict.** Two people or devices may change a file before syncing. The service may merge changes, ask for a choice, or keep conflict copies. Check which version is correct.
- **Deletion travels too.** A deleted or damaged file can sync to other locations. Recycle bins and version history may help, but retention periods and coverage vary.
- **Backup.** A recoverable copy kept under a recovery plan. Sync, sharing, and backup can be features of one product, but their jobs differ. Test restoration of important data.

SCENARIO: A file vanishes from the laptop and cloud after a deletion. Sync may be working exactly as configured. Look for supported recovery options and separate backup copies. [21]

Shared responsibility: protect your part

VISUAL: Draw three bright purple and cyan guards checking identity, opening an allowed door, and writing a record.



The provider has jobs. Your team has jobs too!

GUARD!

- **Provider responsibilities.** The provider protects and operates the service layers it controls. Physical buildings, hardware, and managed software layers depend on the chosen service.
- **Customer responsibilities.** Protect identities and data. Choose safe access and sharing settings. Manage the devices and software layers assigned to your team.
- **IaaS versus SaaS.** With an IaaS VM, the customer normally patches the guest OS. With SaaS, the provider manages the application stack, but customer accounts and data choices still matter.

- **Encryption.** Transforms readable data so it needs the right key to read. In transit means moving between systems. At rest means stored. Use the service's supported protections and protect its keys.
- **Policy / access.** Use approved sign-in methods, least privilege, and review logs. Know which data may enter the service and which recovery features your organization needs.

EXAM TRAP: "It is in the cloud" does not remove customer security duties. Match the responsibility to the service model and actual agreement. [4,5]

Virtual machines: computers inside a computer

VISUAL: Draw one physical server as an apartment house. A purple hypervisor wizard gives resources to three separate OS rooms.



One real machine. Several separate guests!

SPLIT!

- **VM - Virtual Machine.** A software-created computer with virtual hardware. It runs a guest operating system and applications while using physical resources supplied by a host.

- **Host / guest.** The host provides the physical resources. A guest is an OS running in a VM. Multiple guests may share a host, but each VM has its own OS environment.

- **Hypervisor.** Software that creates and runs virtual machines and manages their access to resources. Picture a fair apartment manager dividing space and utilities.

- **Virtual hardware.** A VM can have virtual CPUs, memory, disks, and network adapters. They represent resources; they do not add unlimited real computing power.

- **Use cases.** Run separate services, build practice labs, test software, or support a compatible legacy environment. A broken host can affect many guests at once.

MEMORY HOOK: Physical host = apartment building. Hypervisor = manager. VMs = separate apartments, each with its own OS. Virtualization can exist without cloud computing. [1,22]

Type 1 & Type 2 hypervisors

●●● **VISUAL:** Draw one physical server as an apartment house. A purple hypervisor wizard gives resources to three separate OS rooms.



Look beneath the guests to name the type!

LAYER!

Type	Typical layer order	Common use
Type 1: bare-metal	Hardware → hypervisor → guest VMs	Server consolidation; managed virtual hosts
Type 2: hosted	Hardware → host OS → hypervisor app → guests	Desktop labs and software testing

- **Type 1.** Runs directly on the hardware layer. It may still have a management OS or special management partition. Microsoft Hyper-V is Type 1 even when Windows provides its familiar interface. [22]
- **Type 2.** Runs as an application supported by a host operating system. Oracle VirtualBox is a common hosted example. Features and supported guests depend on the version.

EXAM TRAP: A desktop-looking screen does not prove Type 2. Classify where the hypervisor runs. The management screen is not the whole architecture. [1,22,28]

VM requirements: budget real resources

VISUAL: Draw one physical server as an apartment house. A purple hypervisor wizard gives resources to three separate OS rooms.



Do not give away RAM the host still needs!

PLAN!

- **CPU support.** Check the hypervisor's processor and hardware-virtualization requirements. Intel VT-x means Virtualization Technology for x86; AMD-V means AMD Virtualization. A setting in BIOS or UEFI may need to be enabled.

- **Firmware terms.** BIOS means Basic Input/Output System. UEFI means Unified Extensible Firmware Interface. They initialize hardware before the OS starts. Change settings only as the system guide directs.

- **SLAT.** Second Level Address Translation. Hardware help for translating guest memory addresses. Some hypervisors require it. Check the product rather than assuming all computers qualify.

- **RAM and CPU budget.** Leave capacity for the host and every active guest. A vCPU is a virtual CPU assigned to a VM. Assigning many vCPUs does not create more physical CPU cores.

- **Storage / network / license.** Allow space for guest disks, growth, and snapshots. Provide suitable network access and security controls. Guest OS and app licenses still apply.

EXTRA LAB EXAMPLE: GB means gigabytes. A 16 GB RAM host needs 4 GB for its own work. Two 4 GB guests leave 4 GB of nominal headroom. Overhead and workloads still matter; this is arithmetic practice, not a universal sizing rule. [24]

Virtual disks, snapshots & backups

●●● **VISUAL:** Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



A quick rewind is useful. Keep a real rescue copy!

SAVE!

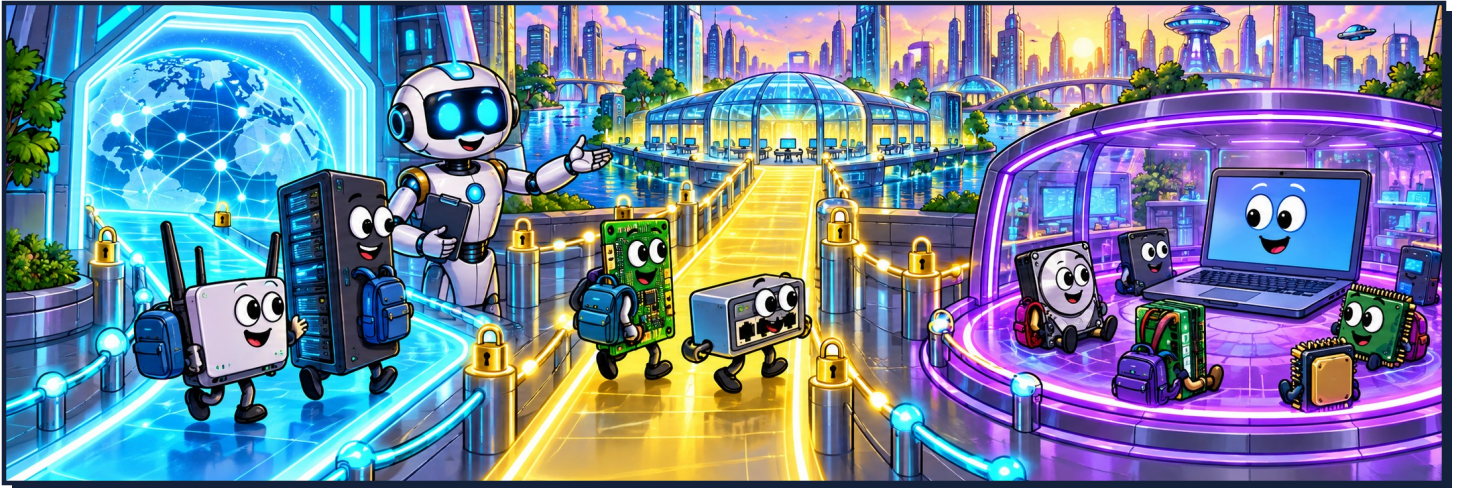
- **Virtual disk.** Storage presented to a VM as a disk. It is often backed by files on host storage. If that storage fills up or fails, the guest can fail too.
- **Thin / dynamic allocation.** A disk file can grow as data is written, up to its configured limit. The advertised virtual capacity is not a promise that the host has enough free physical space.
- **Snapshot / checkpoint.** Records a VM's state at a point in time so supported changes can be rolled back. It may involve disk differences and, depending on type, memory state.

- **Why it is not enough.** A checkpoint may depend on the original disk files. Losing the host storage can destroy both. Keep separate backups and test restores. Standard checkpoints can also have app-consistency limits.
- **Safe use.** Use snapshots for short-term change testing under the product's guidance. Watch growth. Reverting can discard newer work. Never manually delete needed disk-chain files.

EXAM CLUE: "Undo a test change quickly" suggests a snapshot or checkpoint. "Recover after losing the original storage" needs an independent backup. Details and names vary by hypervisor. [25,28]

Virtual networking: choose the path

●●● **VISUAL:** Draw three neon paths: a translator gate, a bridge to the LAN, and an enclosed host-only circle.



Private lab, shared LAN, or translated exit?

CONNECT!

- **vNIC / virtual switch.** A virtual Network Interface Card connects a VM. A virtual switch moves traffic among connected virtual ports and, when configured, a physical network.

- **NAT - Network Address Translation.** A typical NAT mode lets a guest reach outward through translated addresses. Unsolicited inbound access commonly needs extra configuration, such as port forwarding.

- **Bridged mode.** Connects the guest to the host's physical network as a network participant. It needs suitable addressing and follows that network's access rules.

- **Host-only mode.** In VirtualBox, guests can talk to each other and the host on a private virtual network. Outside access is not provided by that mode alone.

- **Internal network.** In VirtualBox, selected guests can share a private network without the host on that network. Product names differ. Added adapters or routing can change the isolation.

LAB RULE: Choose only the access your lab needs. NAT is not a complete security boundary. A VM with another bridged adapter may still reach the real LAN. LAN = Local Area Network. [23]

Sandboxes, legacy apps & test labs

VISUAL: Draw one physical server as an apartment house. A purple hypervisor wizard gives resources to three separate OS rooms.



Try the change in a controlled room first!

TEST!

- **Sandbox.** An isolated place to test behavior with limited effect on other systems. Isolation depends on configuration. Shared folders, clipboards, USB (Universal Serial Bus) devices, and networks can create paths out.

- **Development / testing.** Development builds software. Testing checks its behavior. VMs let a team try different supported OS or app versions without changing each physical computer.

- **Legacy compatibility.** A VM may keep an older OS and app working when the hypervisor supports them. It does not make unsupported software safe, licensed, or compatible with every new CPU.

- **Cross-platform work.** Use another supported OS environment on one computer. CPU architecture matters. Emulation can imitate different hardware; ordinary virtualization does not promise every architecture will run.

- **Application virtualization.** Separates an app from some local system details through packaging, isolation, or remote delivery. It differs from running an entire separate guest OS for every app.

SAFE PRACTICE: Use known, harmless test software and an approved lab. A sandbox is not a reason to run unknown malware on a personal computer. Patch the host and supported guests. [1,22,28]

Containers: apps sharing a kernel

VISUAL: Draw two complete VM capsules with their own engines. Beside them, three app containers share one kernel platform.



Separate app boxes. One shared kernel underneath!

PACK!

- **Container.** An isolated process environment that packages an app with what it needs to run. Containers share the kernel of their host environment.
- **Kernel.** The core part of an OS that manages resources and connects software to hardware. VMs normally have separate guest kernels. Containers on the same container host share one.
- **Image / runtime.** A container image is the template used to create containers. A container runtime starts and manages them. The image is not the same as a running container.

- **Why use them.** They can start quickly and use less overhead than a full guest OS per app. They help teams run consistent app packages across compatible environments.
- **Limits and storage.** Compatibility, security, and persistent data still need planning. Replacing a container must not erase needed records; use appropriate persistent storage. Containers often run inside VMs.

EXAM TRAP: VM = virtual computer with its own guest OS. Container = isolated app environment sharing a kernel. Neither automatically makes an app secure or compatible with every system. [26]

VDI: your desktop runs elsewhere

VISUAL: Draw one physical server as an apartment house. A purple hypervisor wizard gives resources to three separate OS rooms.



Your screen is here. The desktop work is there!

REMOTE!

- **VDI - Virtual Desktop Infrastructure.** Hosts virtual desktop environments centrally. Users connect from endpoint devices to work in those remote desktops.

- **What travels.** The connection carries display updates and user input. The remote system runs the desktop apps. Performance depends on server capacity, connection quality, and the workload.

- **Thin client.** A device built mainly to access remote computing. A full laptop or desktop can also act as the client. The endpoint still needs protection.

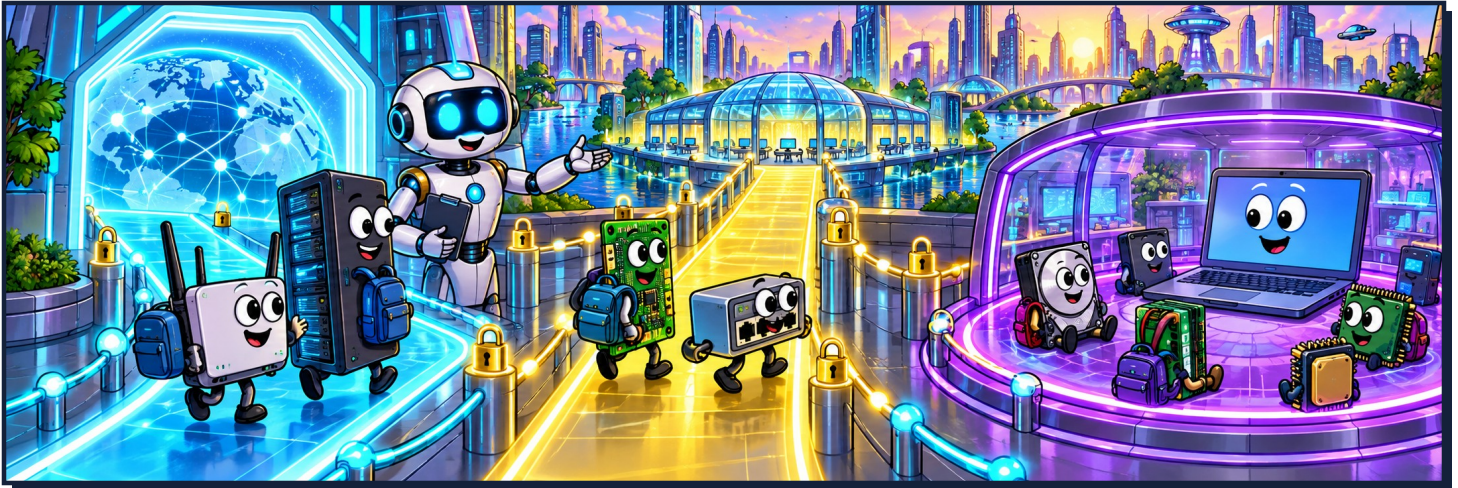
- **Use cases.** A school lab, shared workplace, or remote team can use centrally managed desktops. IT can manage images and access from a common service.

- **Tradeoffs.** Network outages or service problems can interrupt work. Peripheral support, licensing, user profiles, storage, and app needs must fit the solution.

MEMORY HOOK: Local VM = guest runs on your local host. VDI = desktop runs on a remote host and reaches you through a client. DaaS, Desktop as a Service, is a cloud-delivered desktop offering. [1,27]

Troubleshoot the whole chain

VISUAL: Draw three neon paths: a translator gate, a bridge to the LAN, and an enclosed host-only circle.



Follow the clue from the client to the service!

TRACE!

- **1. Identify the problem.** Ask who is affected, when it began, and what changed. Record errors. Protect important work before disruptive tests.
- **2. Test the right layer.** Check the client link and IP settings, DNS, service status, sign-in, and permissions. A server that responds to a ping can still have a broken application.
- **3. Check VM resources.** Confirm the guest is running. Check free host storage, RAM pressure, CPU demand, virtual adapter mode, and guest addressing. Compare healthy guests.
- **4. Check cloud context.** Check service health, region, quotas, billing state, role permissions, and recent changes. Reachable internet does not prove a particular cloud app is healthy.
- **5. Verify and document.** Use a tested plan, make the approved change, confirm the user's actual task works, and record the cause and fix. Escalate when the evidence points outside your access or skill.

SCENARIO: Every guest is slow while the host disk is full. Investigate shared host storage first. Adding cloud capacity or reinstalling every guest would not address the evidence. Extra lab reasoning; use the full CompTIA troubleshooting method in practice. [1]

Your recall mission

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Say it. Explain it. Use it in a scenario!

FLIP!

- **First pass.** Read the lesson, study the illustration, and say the bold terms aloud. For an acronym, say every word before explaining the job.

- **PowerPoint practice.** Open the matching deck in Slide Show mode. Pick a card. Speak your answer before selecting REVEAL ANSWER. Use TRY AGAIN for missed cards.

- **Spaced practice.** Try again later today, tomorrow, and a few days later. Spend extra time on missed cards. Mix server, cloud, and virtualization questions.

- **Printable practice.** The next six pages contain the same 24 cards. Cover the right column. Recall the answer, then uncover it.

- **What the deck does.** Native buttons jump between question and answer slides. No macros are required. It does not save scores or automatically schedule reviews.

MAKE IT STICK: Do more than recite “IaaS.” Say “Infrastructure as a Service,” then explain who patches the guest OS. A short reason turns rote memory into usable understanding.

Flashcards 01-04: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
01. What does a server do? What does a client do?	A server provides a service. A client requests or uses that service.
02. Expand DNS and name its common job.	Domain Name System. It resolves names, commonly to IP addresses.
03. Expand DHCP. What does it give clients?	Dynamic Host Configuration Protocol. It leases IP settings, including options such as gateway and DNS servers.
04. What do file servers and print servers share?	File servers offer network file access. Print servers manage access and queued jobs for shared printers.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Flashcards 05-08: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
05. Which role serves web requests? Which stores organized records?	A web server serves web requests. A database server stores and queries organized data.
06. Which protocol sends mail? Which two retrieve or access it?	SMTP sends or relays mail. IMAP and POP3 let clients access or retrieve mail.
07. Name all three A words and their jobs.	Authentication checks identity. Authorization allows actions. Accounting records activity or use.
08. What does syslog collect? Expand NTP.	Syslog collects event messages. Network Time Protocol keeps clocks in step.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Flashcards 09-12: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
09. Expand IaaS. Who normally patches a rented VM's guest OS?	Infrastructure as a Service. The customer normally patches the guest OS.
10. Expand PaaS. What does the customer mainly bring?	Platform as a Service. The customer brings app code and data to a provider-managed platform.
11. Expand SaaS and give a use case.	Software as a Service. A provider-run email or online document app is an example.
12. Name the four Core 1 cloud deployment models.	Public, private, community, and hybrid.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Flashcards 13-16: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
13. Must a private cloud be on-site? Is public-cloud data public?	No to both. Private clouds can be off-site. Public-cloud data still needs access controls.
14. A service adds capacity for a rush, then removes it. What is this?	Elasticity: capacity adjusts to demand, including shrinking afterward.
15. From a cloud service's view, what are ingress and egress?	Ingress is data entering. Egress is data leaving that service.
16. Why is file sync alone not a complete backup plan?	Deletions or corruption can sync too. Recovery needs retained, recoverable copies and tested restores.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Flashcards 17-20: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
17. Who provides resources? Who runs inside? Who manages VMs?	The host provides resources. The guest OS runs inside a VM. The hypervisor manages virtual machines.
18. How do Type 1 and Type 2 hypervisors differ?	Type 1 runs at the hardware layer. Type 2 runs as an app on a host operating system.
19. Name four resources or requirements to check before adding VMs.	CPU support and capacity, RAM, storage, and network access. Check security and licenses too.
20. Why is a snapshot not a substitute for an independent backup?	It may depend on the original disk files. Losing that storage can lose the snapshot too.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Flashcards 21-24: cover & recall

VISUAL: Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Cover the answer. Speak. Check. Repeat!

FLIP!

Card / question	Answer to recall
21. Which mode translates outbound traffic? Which joins the physical LAN?	Typical NAT mode translates outbound guest traffic. Bridged mode connects the guest to the physical LAN.
22. In VirtualBox, who can communicate in a host-only network?	The host and connected guests. That mode alone does not provide outside-network access.
23. Which normally has its own guest kernel? Which shares a kernel?	A VM has its own guest kernel. Containers share their host environment's kernel.
24. Expand VDI. Where does the desktop run?	Virtual Desktop Infrastructure. The desktop runs on a remote host; the user connects through a client.

Use the matching PowerPoint for clickable answer reveals. Cover the right column for print practice. Revisit the full lesson behind each missed answer.

Sources: servers & cloud models

VISUAL: Draw a smiling cloud above a real data center. Neon cyan server heroes and a purple VM wizard help a student.



Check the facts. Then practice the clues!

READ!

1. CompTIA: 220-1201 exam objectives

[Open primary reference](#)

2. NIST: Definition of cloud computing

[Open primary reference](#)

3. Microsoft: IaaS, PaaS, and SaaS

[Open primary reference](#)

4. Microsoft: Shared cloud responsibility

[Open primary reference](#)

5. AWS: Shared responsibility model

[Open primary reference](#)

6. Microsoft: DNS overview

[Open primary reference](#)

7. Microsoft: DHCP overview

[Open primary reference](#)

8. Microsoft: SMB file sharing

[Open primary reference](#)

9. Cisco: Authentication, authorization, accounting

[Open primary reference](#)

10. IETF: Syslog, RFC 5424

[Open primary reference](#)

Reviewed September 19, 2026. Vendor examples explain concepts; exact features and responsibilities depend on the product and agreement. Independent material; no CompTIA or Per Scholas endorsement.

Sources: standards & operations

VISUAL: Draw a smiling cloud above a real data center. Neon cyan server heroes and a purple VM wizard help a student.



Check the facts. Then practice the clues!

READ!

11. IETF: Network Time Protocol, RFC 5905

[Open primary reference](#)

12. AWS: Load balancing

[Open primary reference](#)

13. CompTIA: Core 1 certification scope

[Open primary reference](#)

14. NIST: Cloud definition publication record

[Open primary reference](#)

15. StarTech: Rack units and mounting

[Open primary reference](#)

16. PostgreSQL: Relational database concepts

[Open primary reference](#)

17. IETF: Email TLS, RFC 8314

[Open primary reference](#)

18. NIST: SCADA definition

[Open primary reference](#)

19. AWS: How pricing works

[Open primary reference](#)

20. Microsoft: Reliability and availability

[Open primary reference](#)

Reviewed September 19, 2026. Vendor examples explain concepts; exact features and responsibilities depend on the product and agreement. Independent material; no CompTIA or Per Scholas endorsement.

Sources: virtualization & recovery

VISUAL: Draw a smiling cloud above a real data center. Neon cyan server heroes and a purple VM wizard help a student.



Check the facts. Then practice the clues!

READ!

21. Microsoft: OneDrive file synchronization

[Open primary reference](#)

22. Microsoft: Hyper-V overview

[Open primary reference](#)

23. Oracle: VirtualBox virtual networking

[Open primary reference](#)

24. Microsoft: Hyper-V hardware requirements

[Open primary reference](#)

25. Microsoft: Hyper-V checkpoints

[Open primary reference](#)

26. Docker: Containers and virtual machines

[Open primary reference](#)

27. Microsoft: Azure Virtual Desktop overview

[Open primary reference](#)

28. Oracle: VirtualBox basics and snapshots

[Open primary reference](#)

Reviewed September 19, 2026. Vendor examples explain concepts; exact features and responsibilities depend on the product and agreement. Independent material; no CompTIA or Per Scholas endorsement.

STAR QUIZ TIME

● ● ● **VISUAL:** Draw a snapshot camera beside a server and a separate backup vault. Add heroes practicing with bright blank flashcards.



Three missions. You have the tools!

YOU GOT IT!

1. A laptop needs automatic IP settings. Later, it must find a website by name. Which two server roles help, in that order? Expand both acronyms.
2. Your team rents a VM and controls its guest OS. Which cloud service model is this? Who normally patches that guest OS?
3. A host drive fails. The VM and its checkpoints were on that same drive. Why might the checkpoints fail to rescue the VM, and what should the recovery plan include?

ANSWERS - CHECK YOUR HERO SKILLS!

1. **DHCP, then DNS.** Dynamic Host Configuration Protocol leases settings. Domain Name System resolves names. Nice work separating the two service desks!
2. **IaaS: Infrastructure as a Service.** The customer normally patches the guest OS. Great job matching control with responsibility!
3. **They may depend on the lost original storage.** Keep independent backups and test restores. You spotted the difference between a quick rewind and a rescue copy!